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Request for Information: Development of an Artificial Intelligence Action Plan

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General Comment

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As an interdisciplinary team of AI researchers, we wish to highlight key considerations for AI in the context of pervasive sensing technologies:

1. Material Foundations of AI: The Role of Sensors in Driving AI Innovation
AI systems are increasingly embedded in consumer products. As the material foundation of many AI systems, sensors directly shape data collection and processing. Since the performance of AI systems heavily depends on data quality, any comprehensive AI Action Plan must account for the role of sensors in AI development and outcomes. A sensor-aware AI framework will provide a strategic advantage that allows the U.S. to develop AI solutions that are both cutting-edge and responsibly designed to minimize downstream liabilities.

2. Calibration and Documentation of Sensor-Based AI Systems for Competitive Advantage
A significant challenge in sensor-driven AI systems is calibration—ensuring that sensors provide reliable data across populations and environments. Miscalibrated sensors can contribute to systemic inaccuracies, leading to faulty outputs. The AI Action Plan should mandate transparent calibration standards and documentation requirements for AI systems relying on sensor data. Establishing high standards for calibration will lead to superior AI performance, cementing U.S. leadership in AI-driven innovation.

3. Mitigating Privacy and Pervasive Sensing Risks as a Catalyst for Market Trust
Many sensors operate in opaque ways, making it difficult for individuals to opt out of data collection. The AI Action Plan should address sensor fusion risks, wherein multiple sensor data streams are combined to infer additional personal information beyond what individuals knowingly disclose. By proactively addressing these risks, the U.S. can lead in creating AI systems that balance innovation with public trust, a necessary component for sustained market adoption and long-term leadership.

4. Proprietary Data Profusion and Market Dynamics Driving AI Leadership
The volume of proprietary sensor data has surged, raising concerns about market concentration and slowed innovation The AI Action Plan should encourage public-interest data infrastructure by promoting open-source sensor datasets and transparent data-sharing agreements. By fostering a collaborative AI ecosystem, where innovation is built on high-quality, accessible sensor data, the U.S. will be able to accelerate technological advancements and solidify its leadership in AI development.

5. Environmental Sustainability of AI-Enabled Sensors as an Innovation Priority
The manufacturing, deployment, and disposal of AI sensors contribute significantly to electronic waste and resource depletion. The AI Action Plan should include guidelines for sustainable sensor design, energy-efficient AI architectures, and responsible e-waste management. Additionally, it should support research into biodegradable and recyclable sensor components to mitigate long-term environmental consequences. A leadership position in sustainable and more efficient AI will not only benefit the planet but also provide U.S. companies with a competitive edge in emerging global AI markets.

Recommendations

• Establishing Independent Calibration and Documentation Standards for Competitive AI Systems: AI models dependent on sensor inputs should include independent validation mechanisms, with publicly accessible documentation to ensure accountability. This will make AI more effective and competitive in the global arena.

• Streamline Sensor Fusion and Data Aggregation Practices: Policymakers should develop guidelines for how multiple data streams from AI-driven sensors can be combined and used for inference. By prioritizing better data aggregation practices for AI, the U.S. will attract both market trust and international leadership opportunities.

• Supporting Energy-Efficient AI Development for Long-Term Market Competitiveness: Federal investment should prioritize research into low-energy AI models, biodegradable sensors, and e-waste reduction strategies to ensure continued AI leadership in sensor-driven AI technology.

• Developing Public Data Commons to Foster AI Innovation: The government should facilitate the creation of open-source sensor data repositories to support rapid innovation. Accessible, high-quality sensor data will drive more breakthroughs, giving U.S. companies a competitive advantage in AI.

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Attachments

NSF Comment Sensors

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March 6, 2025

Subject: Comment on the Development of an AI Action Plan

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To Whom It May Concern,

We appreciate the opportunity to provide comments on the development of the AI Action Plan. As an interdisciplinary team of researchers in the field of AI, with a specific focus on the intersection of AI and pervasive sensing technologies, we wish to highlight several crucial considerations that should be integrated into the AI Action Plan.

Key Considerations

1. Material Foundations of AI: The Role of Sensors in Driving AI Innovation
AI systems are increasingly embedded in consumer products. As the material foundation of many AI systems, sensors directly shape data collection and processing. Since the performance of AI systems heavily depends on data quality, any comprehensive AI Action Plan must account for the role of sensors in AI development and outcomes. A sensor-aware AI framework will provide a strategic advantage that allows the U.S. to develop AI solutions that are both cutting-edge and responsibly designed to minimize downstream liabilities.

2. Calibration and Documentation of Sensor-Based AI Systems for Competitive Advantage
A significant challenge in sensor-driven AI systems is calibration—ensuring that sensors provide reliable data across populations and environments. Miscalibrated sensors can contribute to systemic inaccuracies, leading to faulty outputs. The AI Action Plan should mandate transparent calibration

standards and documentation requirements for AI systems relying on sensor data. Establishing high standards for calibration will lead to superior AI performance, cementing U.S. leadership in AI-driven innovation.

3. Mitigating Privacy and Pervasive Sensing Risks as a Catalyst for Market Trust
Many sensors operate in opaque ways, making it difficult for individuals to opt out of data collection. The AI Action Plan should address sensor fusion risks, wherein multiple sensor data streams are combined to infer additional personal information beyond what individuals knowingly disclose. By proactively addressing these risks, the U.S. can lead in creating AI systems that balance innovation with public trust, a necessary component for sustained market adoption and long-term leadership.

4. Proprietary Data Profusion and Market Dynamics Driving AI Leadership
As sensor networks expand, the volume of proprietary data generated has surged, raising concerns about market concentration and slowed innovation The AI Action Plan should encourage public-interest data infrastructure by promoting open-source sensor datasets and transparent data-sharing agreements. By fostering a collaborative AI ecosystem, where innovation is built on high-quality, accessible sensor data, the U.S. will be able to accelerate technological advancements and solidify its leadership in AI development.

5. Environmental Sustainability of AI-Enabled Sensors as an Innovation Priority
The environmental impact of AI-driven sensors remains underestimated. The manufacturing, deployment, and disposal of AI sensors contribute significantly to electronic waste and resource depletion. The AI Action Plan should include guidelines for sustainable sensor design, energy-efficient AI architectures, and responsible e-waste management. Additionally, it should support research into biodegradable and recyclable sensor components to mitigate long-term environmental consequences. A leadership position in sustainable and more efficient AI will not only benefit the planet but also provide U.S. companies with a competitive edge in emerging global AI markets.

Recommendations

To address the above considerations, we recommend:

- **Establishing Independent Calibration and Documentation Standards for Competitive AI Systems**: AI models dependent on sensor inputs should include independent validation mechanisms, with publicly accessible documentation to ensure accountability. This will make AI more effective and competitive in the global arena.

- **Streamline Sensor Fusion and Data Aggregation Practices**: Policymakers should develop guidelines for how multiple data streams from AI-driven sensors can be combined and used for inference. By prioritizing better data aggregation practices for AI, the U.S. will attract both market trust and international leadership opportunities.

- **Supporting Energy-Efficient AI Development for Long-Term Market Competitiveness**: Federal investment should prioritize research into low-energy AI models, biodegradable sensors, and e-waste reduction strategies to ensure continued AI leadership in sensor-driven AI technology.

- **Developing Public Data Commons to Foster AI Innovation**: The government should facilitate the creation of open-source sensor data repositories to support rapid innovation. Accessible, high-quality sensor data will drive more breakthroughs, giving U.S. companies a competitive advantage in AI**.**

The AI Action Plan presents a crucial opportunity to ensure that AI innovation is precise, reliable, and trustworthy, setting the U.S. apart in global AI markets. This is particularly true for sensor-driven AI. By establishing calibration standards, streamlining sensor fusion and data aggregation practices and supporting sustainable AI development, policymakers can ensure sensor-driven AI is deployed in ways that are a catalyst for innovation.

We appreciate your consideration of these recommendations and welcome further discussions on how to integrate them into AI governance efforts.

Sincerely,

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